Multiple Choice:

1. A pulse modulation technique as the width of a constant amplitude pulse is varied proportional to the amplitude of the analog signal at the time the signal is sampled.
   A. Pulse Width Modulation
   B. Pulse Length Modulation
   C. Pulse Duration Modulation
   D. All of these *

2. It is the transmittal of digitally modulated analog signals (carrier) between two or more points in a communications system.
   A. Digital modulation *
   B. Digital transmission
   C. Data communications
   D. Pulse modulation

3. Indicate which of the following systems is digital.
   A. Pulse-position modulation
   B. Pulse-code modulation *
   C. Pulse–width modulation
   D. Pulse-frequency modulation

4. The baud rate
   A. is always equal to the bit transfer rate
   B. is equal to twice the bandwidth of an ideal channel *
   C. is not equal to the signaling rate
   D. is equal to one – half the bandwidth of an ideal channel

5. Bit errors in data transmission are usually caused by
   A. equipment failures
   B. typing mistakes
   C. noise *
   D. poor S/N ratio at receiver

6. It is a process of converting an infinite number of possibilities to a finite number of conditions.
   A. Sampling
   B. Coding
   C. Quantization *
   D. Aliasing

7. A digital modulation technique which is a form of constant – amplitude angle modulation similar to standard frequency modulation except the modulating signal is binary signal that varies between two discreet voltage levels
   A. QAM
   B. ASK
   C. PSK
   D. FSK *

8. Start and stop bits, respectively, are
   A. Mark, space
   B. Space, mark *
   C. Space, space
   D. Mark, mark

9. A form of angle – modulated, constant amplitude digital modulation similar to conventional phase modulation except its input is binary digital signal and there are limited numbers of output phase possible.
   A. ASK
   B. PSK *
   C. FSK
   D. QAM

10. Refers to the rate of change of a signal on a transmission medium after encoding and modulation have occurred
    A. baud rate *
    B. phase shift
    C. bit rate
    D. frequency deviation

11. The magnitude difference between adjacent steps in quantization is called __________.
    A. Quantum
    B. Step size
    C. Resolution
    D. Any of these *

12. The Hartley – Shannon theorem sets a limit on the
    A. highest frequency that may be sent over a given channel
    B. maximum capacity of a channel with a given noise level *
    C. maximum number of coding levels in a channel with a given noise level
    D. maximum number of quantizing levels in a channel of a given bandwidth

13. It is the ratio of the transmission bit rate to the minimum bandwidth required for a particular modulation scheme.
    A. Bandwidth efficiency
    B. All of these *
    C. Information density
    D. Spectral efficiency

14. Transmitting the data signal directly over the medium is referred to as
    A. baseband *
    B. broadband
    C. ring
    D. bus

15. The main reason that serial transmission is preferred to parallel transmission is that?
    A. serial is faster
    B. serial requires only a single channel
    C. serial requires multiple channels
    D. parallel is too expensive *

16. Any rounded – off errors in the transmitted signal are reproduced when the code is converted back to analog in the receiver.
    A. Aperture error
    B. Quantization error *
    C. Aperture distortion
    D. Slope overload

17. The biggest disadvantage of PCM is
    A. its inability to handle analog signals
    B. the high error rate which is quantizing noise introduces
    C. its incompatibility with TDM
    D. the large bandwidths that are required for it *

18. It is the process of compressing and expanding and is a means of improving the dynamic range of communications system.
    A. Pre-emphasis
    B. Filtering
    C. De-emphasis
    D. Companding *

19. The primary advantage of digital transmission
    A. economical
    B. reliability
    C. noise immunity *
    D. efficiency

20. Part of the PCM system that prevents aliasing or foldover distortion
    A. Bandpass filter
    B. Anti – foldover distortion
    C. Anti – aliasing
    D. Any of these *

21. It is defined as the process of transforming messages or signals in accordance with a definite set of rules.
    A. Quantizing
    B. Sampling
    C. Coding *
    D. Decoding

22. In digital transmission, _____ the levels of a signal may reduce the reliability of the system.

A. Decreasing
B. Maintaining
C. Increasing *
D. Cannot be determined

23. In digital transmission, the _____ is a measure of how fast we can actually send data through a network.
A. Data
B. Output
C. Throughput *
D. None of the above

24. _____ measures the time required for a bit to travel from the source to the destination.
A. Propagation time *
B. Departure time
C. Queuing time
D. Arrival time

25. The _____ is the time needed for each intermediate or end device to hold the message before it can be processed.
A. Propagation time
B. Departure time
C. Queuing time *
D. Arrival time

26. In digital transmission, the ______ defines the number of bits that can fill the link.
A. Link limit
B. Bandwidth-delay product *
C. Link capacity
D. None of the above

27. The ____ defines the number of data elements (bits) sent in 1s. The unit is bits per second (bps).
A. Throughput
B. Bit rate *
C. Output
D. None of the above

28. The ____ is the number of signal elements sent in 1s. The unit is the baud.
A. Baud rate *
B. Bitrate
C. Throughput
D. None of the above

29. According to the Nyquist theorem, the sampling rate must be at least __ times the highest frequency contained in the signal.
A. 1.5
B. 2 *
C. 3
D. 4

30. In data transmission, _____ means reducing the instantaneous voltage amplitude for large values.
A. Encoding
B. Decoding
C. Compression
D. Companding *

31. The last step in PCM is ______.
A. Decoding
B. Companding
C. Encoding *
D. Acceptance

32. Binary data may be organized into groups of n bits each. By grouping, we can send data n bits at a time instead of 1. This is called _____ transmission.
A. Serial
B. Parallel *
C. Asynchronous
D. Direct

33. In ____ transmission, one bit follows another, so we need only one communication channel rather than n to transmit data between two communicating devices.
A. Serial *
B. Parallel
C. Asynchronous
D. Direct

34. In ______ transmission, we send 1 start bit (0) at the beginning and 1 or more stop bits (1s) at the end of each byte. There may be a gap between bytes.
A. Serial
B. Parallel
C. Asynchronous *
D. Synchronous

35. _____ is the dominant method of digital-to-analog modulation.
A. PAM
B. QAM *
C. PSK
D. FSK

36. The total bandwidth required for AM is _____ can be determined from the bandwidth of the audio signal.
A. $B_{AM} = 2B$
B. $B_{AM} = 3B$
C. $B_{AM} = 4B$
D. $B_{AM} = 5B$

37. ______ is the process of changing one of the characteristics of an analog signal based on the information in digital data.
A. Analog conversion
B. Digital conversion
C. Digital-to-analog conversion *
D. Analog-to-digital conversion

38. It is a numerical indication of how efficiently a PCM code is utilized
A. Coding efficiency *
B. Companding
C. Pre-emphasis
D. Dynamic Range

39. Type of PCM which is designed to take advantage of the sample-to-sample redundancies in the typical speech waveform
A. Single – bit PCM code
B. Pulse Code Modulation
C. Differential PCM *
D. Delta modulation

40. The phase relationship between signaling elements for BPSK is the optimum signaling format and occurs only when two binary signal levels are allowed and when one signal is the exact negative of the other.
A. Antipodal signaling *
B. Carrier recovery
C. Squaring loop
D. Phase referencing

41. Pulse – amplitude modulation signals are multiplexed by using
A. Subcarrier
B. Bandpass filters
C. A/D converters
D. FET switches *

42. It is a system where the digital signals are placed directly on the coaxial cable.
A. Broadband
B. Baseband *
C. CSMA/CD
D. Token ring

43. Pulse width modulation may be generated
A. by differentiating pulse position modulation
B. with a monostable multivibrator *
C. by integrating the signal
D. with a free running multivibrator

44. Information capacity is convenient to express as
A. baud
B. bits
C. dot length
D. bits per second or bps *

45. Which medium is the least susceptible to noise?
    A. twin lead
    B. fiber – optic cable *
    C. twisted pair
    D. coax

46. Sixteen different levels (symbols) are used to encode binary data. The channel bandwidth is 36 MHz. The maximum channel capacity is
    A. 18 Mbps
    B. 72 Mbps
    C. 288 Mbps *
    D. 2.176 Gbps

47. Quantizing noise occurs in
    A. time-division multiplex
    B. frequency-division multiplex
    C. pulse-code modulation *
    D. pulse-width modulation

48. In order to reduce quantizing noise, one must
    A. increase the number of standard amplitudes *
    B. send pulses whose sides are more nearly vertical
    C. use an RF amplifier at the receiver
    D. increase the number of samples per second

49. A form of digital modulation similar to PSK except the digital information is contained in both the amplitude and the phase of the transmitted carrier.
    A. ASK
    B. FSK
    C. QAM *
    D. PSK

50. For the 16-PSK and a transmission system with a 10kHz bandwidth, determine the maximum bit rate
    A. 40,000 bps *
    B. 80,000 bps
    C. 20,000 bps
    D. 16,000 bps